

CHM111 Practice Exam on chapter(s): 1, 2

1.  $5.0 \times 10^{-14} \text{ s}$
2.  $8.41 \times 10^1 \text{ Gm}$
3.  $\text{PF}_3$  is a molecular compound.
4. 10 protons and 10 neutrons.
5. Yes, zinc and sulfur form an ionic compound with the empirical formula  $\text{ZnS}$ .
6.  $0.0890 \frac{\text{g}}{\text{mL}}$
7.  $6.5 \times 10^3 \text{ mol}$
8. X is a pure substance; Mass of  $\text{E}_1$ : 4.4 g
9. 0.0082 m
10.  $4.7\% \therefore \% \text{ Error} = \left| \frac{E-A}{A} \right| \times 100\%$
11. 0.000058 m
12.  $4.73 \times 10^8 \text{ W}$
13.  $3.0 \times 10^0 \text{ mm}$
14. 6.9 mL
15.  $30.69 \frac{\text{m}}{\text{s}}$
16.  $77^\circ \text{F}$
17.  $10^{-3}$
18. Carbonic acid
19.  $4.7 \times 10^3 \frac{\text{kg}}{\text{m}^3}$
20. Chromium (III) sulfite:  $\text{Cr}_2(\text{SO}_3)_3$
21.  $1.0 \times 10^2 \text{ g}$
22.  $\text{NH}_3$
23. Copper most commonly forms the cation  $\text{Cu}^+$ .
24.  $\text{Ba}^{2+}$  is a cation. It has 56 protons and 54 electrons.
25. 3.1 mg
26. 82.33 MPa
27.  $1.6 \frac{\text{mol}}{\text{L}}$
28.  $4.0721 \times 10^3 \text{ m}$

29. Systematic error  $\because$  every measurement is consistently too high by the same amount due to a calibration flaw.
30. 265.09 nm
31. X is a pure substance; Mass of  $E_1$ : 4.4 g
32. Physical property. The density of lead is a physical property; it can be measured without changing the chemical identity of lead.
33. The SI base unit for amount of substance is the mole (symbol: mol).
34. Element
35. 0.7469 cm
36. Aluminum sulfate:  $Al_2(SO_4)_3$
37. 70.5 fm
38. 0.56 m
39. 354 K
40. Pure substances: A, B, C